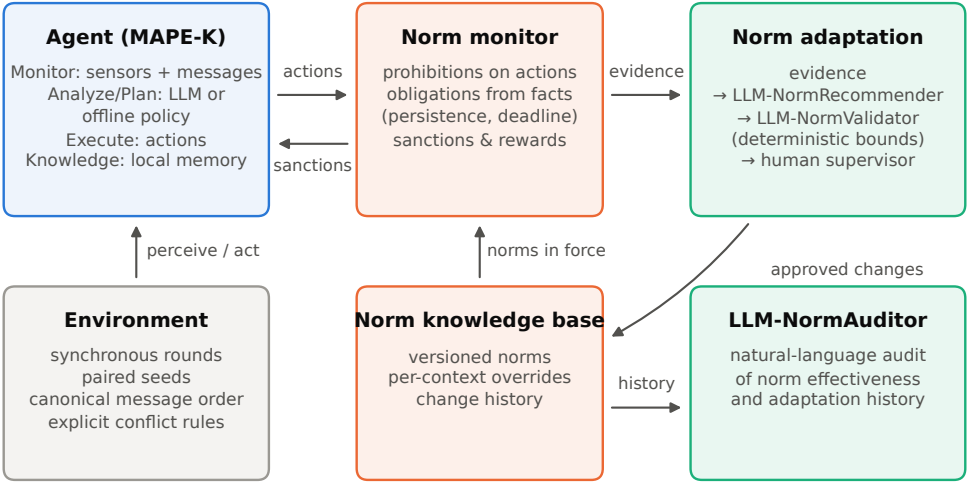
Agent (MAPE-K)
Monitor: sensors + messages
Analyze/Plan: LLM or offline policy
Execute: actions
Knowledge: local memory
actions
sanctions
Norm monitor
prohibitions on actions
obligations from facts (persistence, deadline)
sanctions & rewards
evidence
Norm adaptation
evidence
→ LLM-NormRecommender
→ LLM-NormValidator (deterministic bounds)
→ human supervisor
perceive / act
norms in force
approved changes
Environment
synchronous rounds
paired seeds
canonical message order
explicit conflict rules
Norm knowledge base
versioned norms
per-context overrides
change history
history
LLM-NormAuditor
natural-language audit of norm effectiveness and adaptation history